ROYYA CHAKRAVARTHI

Digital Systems Engineer | RTL Design & Verification | FPGA & ASIC

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SPECIALIZED SUMMARY

M.E. in Digital Systems scholar at University College of Engineering, Osmania University, specializing in RTL design, hardware accelerator architectures, and design verification. Strong hands-on competence in synthesizable Verilog HDL, Cadence Xcelium simulation, INT8 quantization flow, FSM datapaths, and RF electronics testing. Passionate about architecting energy-efficient silicon computing blocks and high-performance digital systems.

EDUCATION

University College of Engineering, Osmania University — Hyderabad, India
Master of Engineering (M.E.) in Digital Systems Pursuing (Expected Aug 2027)
Key Focus: Advanced Digital System Design, RTL Modeling, Low-Power VLSI, Computer Architecture, Verification.

Anurag University — Hyderabad, India
Bachelor of Technology (B.Tech) in ECE 2020 – 2024 | CGPA: 8.58 / 10

VLSI & HARDWARE SKILLS

Hardware Description & RTL: Verilog HDL, SystemVerilog (Fundamentals), Synthesizable RTL, FSM Datapath Design, Pipelining

EDA & Simulation Tools: Cadence Xcelium, SimVision Waveform Viewer, Yosys (OpenLane flow study), ModelSim, Linux EDA environment

Hardware AI & Verification: INT8 Fixed-Point Quantization, Systolic MAC Arrays, Self-Checking Testbenches, Waveform Debugging

Supporting Languages & Tools: C Programming, Python (NumPy/TensorFlow for tensor verification), MATLAB, Git, Shell Scripting

ENGINEERING EXPERIENCE

Astra Microwave Products Limited — Hyderabad, India
Project Trainee February 2024 – June 2024 [Project Trainee]
- Supported testing, RF characterization, and diagnostic analysis of aerospace & defense-grade microwave components.
- Operated high-frequency laboratory test equipment including Spectrum Analyzers, Signal Generators, and RF Power Meters.
- Evaluated PCB signal paths and documented RF attenuation and insertion loss data against defense specifications.

HARDWARE & RTL PROJECTS

Hardware Accelerator Design of LeNet-5 CNN | Verilog HDL, Cadence Xcelium, Python
- Engineered synthesizable RTL architecture for LeNet-5 CNN inference using INT8 symmetric quantization.
- Designed modular convolutional line buffers and pipelined Multiply-Accumulate (MAC) arrays in Verilog HDL.
- Conducted cycle-accurate verification in Cadence Xcelium, confirming complete numerical parity with software gold tensors.

RTL 32-Bit ALU Design & Verification | Verilog HDL, Cadence SimVision
- Modeled parameterized 32-bit ALU with 16 distinct arithmetic, logical, and barrel-shift instructions.
- Developed comprehensive self-checking testbenches validating Zero, Carry, Sign, and Overflow flag generation.

Delta-Sigma Modulator (DSM) Digital Decimation Filter | Academic Research & MATLAB Modeling
- Formulated multi-stage decimation architectures featuring multiplier-free Cascaded Integrator-Comb (CIC) and half-band FIR filters.